

▼ BERT Misclassification Analysis

```
!pip install -q kaggle
from google.colab import files
files.upload()
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
!kaggle datasets download -d nikhileswarkomati/suicide-watch
!unzip suicide-watch.zip
import pandas as pd
df = pd.read_csv('Suicide_Detection.csv')
print(f"Dataset shape: {df.shape}")
print(df.head())
```

Choose Files kaggle.json

- **kaggle.json**(application/json) - 69 bytes, last modified: 4/6/2025 - 100% done

Saving kaggle.json to kaggle.json
 Dataset URL: <https://www.kaggle.com/datasets/nikhileswarkomati/suicide-watch>
 License(s): CC-BY-SA-4.0
 Archive: suicide-watch.zip
 Inflating: Suicide_Detection.csv
 Dataset shape: (232074, 3)

Unnamed: 0	text	class
0	2 Ex Wife Threatening SuicideRecently I left my ...	suicide
1	3 Am I weird I don't get affected by compliments...	non-suicide
2	4 Finally 2020 is almost over... So I can never ...	non-suicide
3	8 i need helpjust help me im crying so hard	suicide
4	9 I'm so lostHello, my name is Adam (16) and I've...	suicide

```
import pandas as pd
import torch
from torch.utils.data import Dataset, DataLoader
from transformers import AutoTokenizer, AutoModelForSequenceClassification, get_linear_schedule_with_warmup
from torch.optim import AdamW
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from tqdm.auto import tqdm
import random
```

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
df = pd.read_csv("Suicide_Detection.csv").dropna()
df['label'] = df['class'].map({'non-suicide': 0, 'suicide': 1})
df_small = df.groupby('label', group_keys=False).apply(lambda x: x.sample(n=5000, random_state=42)).reset_index(drop=True)
```

```
train_texts, val_texts, train_labels, val_labels = train_test_split(
    df_small['text'].tolist(), df_small['label'].tolist(), test_size=0.2, stratify=df_small['label'], random_state=42
)
```

```
tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
train_enc = tokenizer(train_texts, truncation=True, padding=True, max_length=128)
val_enc = tokenizer(val_texts, truncation=True, padding=True, max_length=128)
```

```
class RedditDataset(Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels
    def __getitem__(self, idx):
        item = {k: torch.tensor(v[idx]) for k, v in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels[idx])
        return item
    def __len__(self):
        return len(self.labels)
```

```
train_ds = RedditDataset(train_enc, train_labels)
val_ds = RedditDataset(val_enc, val_labels)
train_loader = DataLoader(train_ds, batch_size=8, shuffle=True)
val_loader = DataLoader(val_ds, batch_size=8)
```

```
model = AutoModelForSequenceClassification.from_pretrained("bert-base-uncased", num_labels=2)
model.to(device)
optimizer = AdamW(model.parameters(), lr=2e-5)
scheduler = get_linear_schedule_with_warmup(optimizer, num_warmup_steps=0, num_training_steps=len(train_loader))
```

```
model.train()
for batch in tqdm(train_loader, desc="Training Epoch 1"):
    optimizer.zero_grad()
    batch = {k: v.to(device) for k, v in batch.items()}
    outputs = model(**batch)
    loss = outputs.loss
    loss.backward()
    torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
    optimizer.step()
    scheduler.step()
```

```
model.eval()
preds, true = [], []
with torch.no_grad():
    for batch in val_loader:
        batch = {k: v.to(device) for k, v in batch.items()}
        outputs = model(**batch)
        logits = outputs.logits
        preds.extend(torch.argmax(logits, dim=1).cpu().numpy())
        true.extend(batch['labels'].cpu().numpy())
```

```
false_positives = [val_texts[i] for i in range(len(true)) if preds[i] == 1 and true[i] == 0]
false_negatives = [val_texts[i] for i in range(len(true)) if preds[i] == 0 and true[i] == 1]
```

```
print("\nFalse Positives (predicted suicide, actually non-suicide):\n")
for x in random.sample(false_positives, min(5, len(false_positives))): print("-", x, "\n")
print("\nFalse Negatives (missed actual suicidal posts):\n")
for x in random.sample(false_negatives, min(5, len(false_negatives))): print("-", x, "\n")
```

```
<ipython-input-3-abc342188f9>:4: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the
df_small = df.groupby('label', group_keys=False).apply(lambda x: x.sample(n=5000, random_state=42)).reset_index(drop=True)

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 886B/s]
config.json: 100% 570/570 [00:00<00:00, 29.0kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 3.05MB/s]
tokenizer.json: 100% 466k/466k [00:00<00:00, 7.30MB/s]
Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better performance, install the package with: `pip in
WARNING:huggingface_hub.file_download:Xet Storage is enabled for this repo, but the 'hf_xet' package is not installed. Falling back to regular HTTP download. For better perform
model.safetensors: 100% 440M/440M [00:01<00:00, 259MB/s]
Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-uncased and are newly initialized: ['classifier.bias', 'classifier.weig
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Training Epoch 1: 100% 1000/1000 [03:10<00:00, 5.16it/s]

False Positives (predicted suicide, actually non-suicide):

- Hhhhhhhh im tired of this Trigger warning because i dont wanna hurt people or make them have a panic attack, this includes mention of self harm and other things that may possi

So this another self check because I haven't done one in like a while because i was doing perfectly fine. Key word was. Lets start with the first thing, i decided to harm myself

- How can I fix this? I keep passing out on the bus in the morning even though when I get to the stop I am energized and am almost passing out during classes?

- A possible suicide happened when I was going home from school Just to be clear I live in japan and it's 6:28 as I write this. I am in a bus and there might be typo's cause I

I was going home from school when my teacher said that there was a (accident caused by human) happened. It was on a nearby train station and most of our students ride the
I don't know who it is or what they did, but I'm pretty sure it was suicide. My friend said he saw blood on the rails on the track and an ambulance came along with like 50 polic
I'm posting here because teenagers are more likely to commit suicide (cause you know, hormones going crazy, puberty, etc) I simply want to raise awareness. So please, no awards.

I'll update you guys as time follows. Bye for now

- Honestly just fuck Covid Bit of a story here, but I just wanted to say it. All my life I've wanted to travel around. A good way to do this was to get into my schools Chinese c

I finally got in my freshman year, but the trip was scheduled for April 2020. So needless to say it got cancelled. I was sad, but they said that next year we'd have a chance to

- Does anyone experience this too? Sometimes when my day was extremely exhausting and a lot of noise was going on and I finally go to bed and close my eyes, there seem to be the

False Negatives (missed actual suicidal posts):

- My friend is obsessed with his exOK I (20F) have a friend (20M) that has so much issues and I just don't know what to do anymore. I'm just going to explain his story because I

So this friend, lets call him Rick, has had a really rough year with his mom dying and a boyfriend (yeah he gay) that cheated while they we're 'on a break'. Rachel and Ross kind

So the situation is my friend, that now has a broken hand, no job, no boyfriend, no mom, no school, and no home.

I still think this ex is a dick for cheating on him, when his mom just f*cking past away for like a month and all of the other circumstances. I have to say that they we're on a

Rick still cries over him everyday and really misses him because he was the only good thing he had when his mom died. Every time I come home from work and ask how he's doing, ar

I already tried talking to Rick about that he really has to calm down on this and have to respect the decision of the ex about that he (clearly) want's to be left alone. He has

I told him to talk to a psychiatrist and he went to one meeting. He said that it was OK but not really convinced.

The only thing he's talking about is his ex, he doesn't care about his own life but does care about his vanity, HE BUYS A LOT OF STUFF, everyday I come home he bought something

I'm really starting to feel a bit irritated because he won't listed to me, he won't look for a job and he's living with me since the end of November, takes all the money from hi

Does anyone have any tips maybe... I feel like this is a bit of a stretch because I already asked friends and everyone says things like: If he doesn't want te be helped, there i

Sorry for my English BTW.

- Am I allowed to feel even the slightest bit happy ?I feel as if the moment I relax myself even a little or let my guard down the slightest or feel the smallest bit of happines

- guess this is me writing..a little background about me, i'm 17M, from down south in the US, about to be a senior in high school.

uhhh I don't really know what to write so i'm just gonna ramble and any advice or comments or whatever comes to mind, you can leave below:)

recently i've been falling back down into my rabbit hole of depression and another episode has started and they get worse and worse each time. normally these episodes are either

one big issue i've been facing is rejection and being tossed aside. people love to lead me on until I am convinced something good is happening and then when that happens, they l

i've become really apathetic about really everything, about friends caring if I were gone or not, and the only thing that shakes me to my core about being gone would be my famil

thanks to anyone who read this far, I really appreciate it. like I said I just needed to ramble and get that kinda stuff out to someone other than the usual 2-3 people that hear

- :(i literally cant remember the last time i had a good day. it's been so fcking long and every week, every month it just gets worse, like i cant remember the last time i thoug

- Good bye everyone.I'm not gonna bother writing about the things in my life. I just want to say good bye to someone, as it feels like the right thing to do. So good bye people.
```

```
print("\nClassification Report:\n")
print(classification_report(true, preds, target_names=['Non-suicide', 'Suicide']))
```

Classification Report:

	precision	recall	f1-score	support
Non-suicide	0.97	0.96	0.97	1000
Suicide	0.96	0.97	0.97	1000
accuracy			0.97	2000
macro avg	0.97	0.97	0.97	2000
weighted avg	0.97	0.97	0.97	2000